Dataset Details & Specification

Open University Learning Analytics Dataset (OULAD) Documentation

1. Dataset Overview & Provenance

- **Dataset Name:** Open University Learning Analytics Dataset (OULAD)
- **Source:** Open University (United Kingdom), Knowledge Media Institute (KMi)
- **Primary Link:** https://analyse.kmi.open.ac.uk/open_dataset
- **Alternative Figshare DOI:** https://doi.org/10.6084/m9.figshare.880490
- **Format:** Relational CSV (7 tables)
- **Total Volume:** 32,593 student enrolments, 22 module presentations, 10.6 million clickstream interactions

2. Dataset Purpose

OULAD was released to support learning analytics research on student demographics, study trajectories, virtual learning environment (VLE) engagement, and assessment outcomes. Its granular clickstream logs and assessment records make it the premier benchmark for early-warning predictive modeling, student prioritization, and personalized academic intervention.

3. Relational Table Structure & Key Fields

All 22 key fields specified in the Capstone guidelines are incorporated and mapped across the 7 relational tables:

Field Name	Source Table	Data Type & Description
id_student	studentInfo	Unique anonymized integer student identifier
code_module	courses, studentInfo	Course module code (AAA through GGG)
code_presentation	courses, studentInfo	Presentation code (e.g. 2013B, 2013J, 2014B, 2014J)
gender	studentInfo	M / F (Sensitive attribute; audit only)
region	studentInfo	13 UK geographic regions (Sensitive; audit only)
highest_education	studentInfo	Highest prior qualification level
imd_band	studentInfo	Index of Multiple Deprivation decile band
age_band	studentInfo	Age bracket: 0-35, 35-55, 55<=
num_of_prev_attempts	studentInfo	Number of times student previously attempted this module
studied_credits	studentInfo	Total credits currently being studied (workload)
final_result	studentInfo	Final outcome: Distinction, Pass, Fail, Withdrawn
id_assessment	assessments	Unique assessment identifier
date_submitted	studentAssessment	Day of submission relative to course start date
score	studentAssessment	Assessment score (0–100)
date_registration	studentRegistration	Day student enrolled relative to course start
date_unregistration	studentRegistration	Day student cancelled enrolment (null if completed)
id_site	vle, studentVle	Unique VLE material / resource identifier
date	studentVle	Day interaction occurred relative to course start

sum_click	studentVle	Number of student clicks on resource that day
activity_type	vle	Type: forumng, oucontent, quiz, resource, url, etc.
week_from / week_to	vle	Planned teaching schedule week boundaries

4. Data Preprocessing Performed

1. **Temporal Cutoff Slicing:** Snapshots taken at 15%, 25%, 40%, and 60% of planned course length (approx weeks 6, 9, 15, 22). Any interaction or submission after the cutoff is strictly discarded.
2. **In-Scope Population Filtering:** Students whose date_unregistration is prior to the cutoff are flagged as already withdrawn and excluded from model training, ranking, and retrieval to prevent trivial leakage.
3. **Dense Weekly Engagement Grids:** Aggregated clicks into weekly matrices, computing slope, momentum, and inactivity streaks.
4. **Assessment Scoring Alignment:** Missing assessment scores stay missing (NaN) with explicit no_scores_yet flags.
5. **Sensitive Attribute Isolation:** Protected fields are removed from model feature matrices and agent prompts.